Research Engineer Intern

3 days from receipt, including reading and practical work.

01 Read and discuss

Read both papers. There is nothing to submit for this question. We will discuss them and your practical work in a follow-up interview of approximately 60 minutes.

3D Gaussian Splatting for Real-Time Radiance Field Rendering
Kerbl et al. | SIGGRAPH 2023

Hybrid Particle-Voxel Dynamics for Multimodal 3D World Simulation
Wang et al. | arXiv, 2026

Be ready to explain the main ideas, assumptions, and limitations. You do not need to reproduce either paper.

02 Learn a deformable object from video

Choose **one soft object** and film yourself pressing or squishing it, then letting go. Use the observed motion to build a simple 3D model that simulates its deformation and recovery.

The core task is to **fit the model's behavior from the video**, for example its effective stiffness and damping. Then demonstrate its response to a fresh instance of the same interaction, using the fitted model, and briefly compare the simulation with the real behavior.

Choose your own capture setup, representation, solver, and fitting approach. Explain the assumptions you make and what your model can and cannot capture.

3D reconstruction is not assessed. A rough mesh or simple geometric proxy is fine. You may use existing reconstruction pipelines, pretrained models, simulation libraries, and code. The result should be a deformation model informed by your observations, rather than a replay of the recorded motion.

Share your work

03 Submission

Reply to the assignment email with your **GitHub repository** and a **link to view the result**. A private repository is fine; give the hiring team access.

Include:

- Your code and model, with a README explaining how to run them.
- Your input video, or an accessible link if the files are large.
- A short demo video or interactive viewer showing the real object and its simulated response.
- A brief explanation in the README of your approach, assumptions, results, and limitations. Credit the tools and code you used.

How we evaluate

We care about your technical understanding, how visual observations inform the deformation model, and whether we can reproduce the result. Clear reasoning and an honest explanation of limitations matter more than visual polish.